A2 · Unit 5 · WBI15 Respiration, Internal Environment, Coordination and Gene Technology

Cellular Respiration

Cellular respiration is how a cell releases the chemical energy stored in food and captures it as ATP. This page follows the route of glucose through glycolysis, the link reaction, the Krebs cycle and the electron transport chain, then looks at what changes when oxygen runs out, the respiratory quotient and the two respiration core practicals.

SPEC 7.1 Aerobic respiration: the overall reaction

WHAT THIS SPEC POINT ASKS

(i) Know that aerobic respiration splits the respiratory substrate, releasing CO_2 as a waste product, and joins the hydrogen from it to oxygen, releasing a large amount of energy. (ii) Know that respiration happens in many small steps, each catalysed by its own intracellular enzyme. You don't need enzyme names.

The overall equation

$$C_6H_{12}O_6 + 6O_2 \rightarrow 6CO_2 + 6H_2O \text{ (about 2870 kJ released per mole of glucose)}$$

Read it the way the spec does:

- **Splitting the substrate.** The carbon skeleton of glucose is broken down step by step. Its carbon atoms leave as **CO_2**, a waste product, in the decarboxylation reactions of the link reaction and Krebs cycle.
- **Reuniting hydrogen with oxygen.** Hydrogen atoms are removed from the substrate (dehydrogenation) and carried by the coenzymes **NAD** and **FAD** to the electron transport chain. There they finally combine with **oxygen** from the air to make **water**. This is where most of the energy is released.
- The energy isn't "produced". It is **released** from the substrate and **transferred** to make ATP from ADP + Pi. Much of it (about 60%) is released as **heat**, which helps keep mammals warm.

Oxidation and reduction

	Oxidation	Reduction
Hydrogen	lost	gained
Electrons	lost	gained
Oxygen	gained	lost

- Glucose is **oxidised** (it loses hydrogen). Oxygen is **reduced** (it gains hydrogen and electrons) to water.

- NAD + 2H → reduced NAD. Reduced NAD is the carrier "loaded" with hydrogen; it becomes NAD again when it hands the hydrogen on.

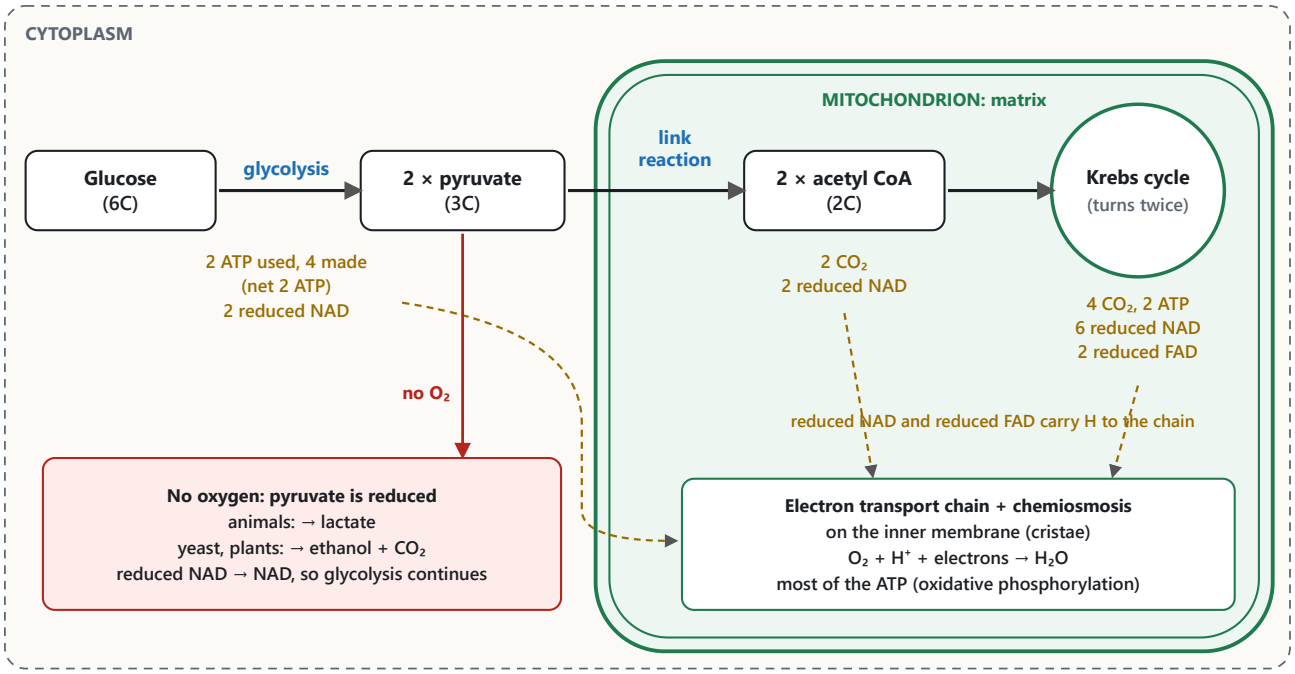
Why respiration happens in steps

- Respiration is a **series of reactions**: the product of one step is the substrate of the next. Each step is catalysed by a **specific intracellular enzyme**, because each enzyme's active site fits only one substrate.
- Releasing the energy in **small amounts**:
  - lets much of it be captured in ATP at several points, instead of all being lost as heat at once (as when glucose burns),
  - avoids a sudden temperature rise that would denature enzymes,
  - allows control: the rate of a key step can be slowed when ATP is plentiful,
  - provides **intermediates** that other pathways use, and entry points for other substrates (fatty acids, amino acids, glycerol).

ATP: the energy currency

- **ATP** (adenosine triphosphate) = adenine + ribose + three phosphate groups.
- **Hydrolysis** of ATP to ADP + Pi, catalysed by **ATPase**, releases energy in small, usable amounts exactly where it's needed: active transport, muscle contraction, protein synthesis, the Na^+/K^+ pump in neurones.
- **ATP synthase** makes ATP from ADP + Pi (a condensation reaction) using energy released in respiration.
- ATP is soluble, moves easily around the cell and is remade very quickly. It is an immediate energy source, not a long-term store.

The four stages at a glance



Aerobic respiration of one glucose molecule. Glycolysis happens in the cytoplasm; the link reaction and Krebs cycle in the mitochondrial matrix; the electron transport chain on the inner membrane. Without oxygen, only glycolysis continues.

Stage	Where	Needs O_2?	Main products per glucose
Glycolysis	cytoplasm	no	2 pyruvate, net 2 ATP, 2 reduced NAD
Link reaction	mitochondrial matrix	yes (indirectly)	2 acetyl CoA, 2 CO_2, 2 reduced NAD
Krebs cycle	mitochondrial matrix	yes (indirectly)	4 CO_2, 2 ATP, 6 reduced NAD, 2 reduced FAD
Oxidative phosphorylation	inner mitochondrial membrane	yes	water, most of the ATP

WORKED EXAMPLE

Worked example: efficiency of aerobic respiration

The question gives the figures: 1 mol glucose releases 2867.48 kJ; up to 38 ATP are made per glucose; hydrolysis of 1 mol ATP releases 30.51 kJ.

Energy captured in ATP = 38 × 30.51 = **1159.38 kJ**.

Efficiency = 1159.38 ÷ 2867.48 × 100 = **40.43%** (WBI15 Jun 2021).

The other energy is released as **heat**.

Same method for a lipid: 122 × 30.51 = 3722.22 kJ; 3722.22 ÷ 12 244 × 100 = **30.4%** to 3 s.f. (WBI15 Jan 2024). "30.40%" was marked wrong: give exactly the s.f. asked for.

MARK-SCHEME WORDING

- Equation: $C_6H_{12}O_6 + 6O_2 \rightarrow 6CO_2 + 6H_2O$. One mark for the correct formulae, one for balancing. ATP/ADP in the equation are ignored. A correct word equation is accepted.
- ATP use: "**hydrolysis** of ATP (to ADP and Pi)" that "**releases energy**", then name the process it powers (active transport, muscle contraction, protein synthesis).
- Energy not trapped in ATP: "lost as **heat**", "raises body temperature".
- Oxygen is "the **final electron acceptor**"; water forms by the **reduction** of oxygen.

Often asked as: write and balance the equation (for glucose or a given lipid); describe how ATP supplies energy; efficiency calculations from supplied data; which molecule first contains labelled oxygen.

COMMON MISTAKE

- "Energy is **produced**." Energy is released or transferred; ATP is produced.
- Writing glucose as $C_6H_{12}O_2$, balancing by putting a 6 in front of glucose, or drawing the arrow backwards.
- A GCSE-level ATP answer ("ATP gives energy for life processes"). Name the process and link it to **hydrolysis** of ATP.
- Thinking the oxygen you breathe in ends up in CO_2. It ends up in **water**, so labelled O_2 first appears in water (WBI15 Oct 2022).

⚠ Beyond the spec · not part of the Edexcel IAL specification

Examined in past papers even though the spec doesn't list it.

- Balancing a lipid equation: $C_{57}H_{104}O_6 + 80O_2 \rightarrow 57CO_2 + 52H_2O$ (WBI15 Jun 2021). Balance C first (gives the CO_2), then H (gives the H_2O), then O last.
- "Stepped and enzyme-controlled" is tested through **inhibitors** that block one step (cyanide, amytal, hydramethylnon), not through enzyme names (WBI15 Oct 2021, Jan 2025). See 7.4.

SPEC 7.2 Glycolysis

WHAT THIS SPEC POINT ASKS

Understand what glycolysis does in both aerobic and anaerobic respiration: phosphorylation of a hexose, ATP made by substrate-level phosphorylation, reduced NAD, pyruvate and (in animals without oxygen) lactate. You don't need the intermediate stages or compounds.

Where and when

- Glycolysis happens in the **cytoplasm**. It doesn't need oxygen, so it is the first stage of **both** aerobic and anaerobic respiration.
- Glyco-lysis means "sugar splitting": one 6C glucose becomes two 3C pyruvate.

The steps (per glucose)

1. **Phosphorylation.** Two ATP are hydrolysed and their phosphate groups are added to glucose, making a phosphorylated hexose (hexose bisphosphate). This makes the sugar more **reactive** (lowers the activation energy for splitting it), and the charged phosphates stop it leaving the cell.
2. **Splitting.** The phosphorylated hexose splits into **two triose phosphate** molecules (3C).
3. **Oxidation.** Hydrogen is removed from each triose phosphate (dehydrogenation). **NAD** picks it up, making **2 reduced NAD**.
4. **ATP formation.** Phosphate groups are transferred from the intermediates directly to ADP, making **4 ATP** by **substrate-level phosphorylation**. The end product is **2 pyruvate** (3C).

KEY IDEA

Net yield per glucose: **2 ATP** (4 made − 2 used), **2 reduced NAD**, **2 pyruvate**. All of this happens without oxygen.

- **Substrate-level phosphorylation**: ATP made by transferring a phosphate group **directly** from a substrate molecule to ADP. It happens in glycolysis and the Krebs cycle and needs no electron transport chain. (Compare oxidative phosphorylation, 7.4.)

What happens to pyruvate next

	Oxygen available	No oxygen (anaerobic)
Animals	actively transported into the mitochondrial **matrix** → link reaction	**reduced to lactate** by reduced NAD (one step, no CO_2)

	Oxygen available	No oxygen (anaerobic)
Yeast and plants	as in animals	decarboxylated to ethanal (**CO_2 released**), then ethanal reduced to **ethanol** by reduced NAD

Why anaerobic respiration makes lactate (or ethanol)

- Without oxygen, the electron transport chain stops, so reduced NAD can't be reoxidised there.
- A cell has only a little NAD. Once it is all reduced, step 3 of glycolysis can't happen and glycolysis would stop.
- Turning pyruvate into lactate uses the hydrogen from reduced NAD, so **NAD is regenerated**. Glycolysis can keep going and keep making **2 ATP per glucose**.
- That is far less ATP than aerobic respiration, because pyruvate isn't oxidised any further: most of the energy stays locked in lactate (or ethanol).

MARK-SCHEME WORDING

- Pyruvate in anaerobic conditions: "**reduced / converted to lactate**"; "reduced NAD is **oxidised**", so NAD is regenerated and "**glycolysis can continue**"; "lactate transported to the liver". ACCEPT "H atoms added to pyruvate".
- Products of glycolysis: "pyruvate", "reduced NAD", "ATP" (net 2).
- Why cyanide doesn't stop anaerobic ATP production: "anaerobic respiration involves just **glycolysis**", "ATP made in glycolysis (by **substrate-level phosphorylation**)", "in the **cytoplasm**, so mitochondria / the ETC aren't needed".
- Less ATP: anaerobic respiration must be linked to **less ATP per glucose**. "Anaerobic respiration" on its own is not credited.

Often asked as: MCQs on where glycolysis happens and on its first step (phosphorylation of hexose); identify ADP, ATP, NAD and reduced NAD on a pathway diagram; why cyanide doesn't affect anaerobic ATP production; why a cheetah can sprint only briefly.

COMMON MISTAKE

- "Anaerobic respiration produces no ATP." It makes a net **2 ATP** per glucose.
- "Lactate is made to produce ATP." Its job is to **regenerate NAD**; the ATP comes from glycolysis.
- Saying CO_2 is released when lactate forms. Only the ethanol pathway releases CO_2.
- Writing NADP or NADPH (those belong to photosynthesis). Write "reduced NAD" to match the spec.
- "The phosphate comes from pyruvate." It comes from the phosphorylated intermediates.

⚠ Beyond the spec · not part of the Edexcel IAL specification

Examined in past papers even though the spec doesn't list it.

- ATP yields are supplied in questions, e.g. anaerobic 2 v aerobic 36 per glucose (WBI15 Jun 2023). Glucose released from **glycogen** is already phosphorylated without using ATP, so it gives **3 ATP** and 2 lactate (WBI15 Oct 2021).
- Cheetah sprints: fast-twitch fibres, anaerobic respiration, lactate build-up and oxygen debt; also heat builds up faster than homeostasis can remove it (WBI15 Oct 2022).
- Log-scale graphs (lactate against body mass in crocodiles). Say aerobic respiration "cannot keep up" with ATP demand in vigorous exercise, and link every trend to aerobic or anaerobic respiration (WBI15 Oct 2021, 6-mark).

SPEC 7.3 Link reaction and Krebs cycle

WHAT THIS SPEC POINT ASKS

Understand how the link reaction and Krebs cycle complete the oxidation of glucose: CO_2 formed by decarboxylation, ATP by substrate-level phosphorylation, and reduced NAD and reduced FAD by dehydrogenation. Both happen in the mitochondria (matrix), while glycolysis happens in the cytoplasm. You don't need the names of other compounds.

Mitochondrion structure

Part	Feature	Role
Outer membrane	smooth	pyruvate is actively transported across it
Inner membrane	folded into **cristae** (large surface area)	holds the electron carriers and **ATP synthase**: oxidative phosphorylation
Intermembrane space	narrow	H^+ builds up here
Matrix	fluid containing enzymes, mitochondrial DNA and ribosomes	**link reaction** and **Krebs cycle**

Three kinds of reaction to recognise

- **Decarboxylation**: a carboxyl group is removed as **CO_2**.
- **Dehydrogenation**: hydrogen is removed (an oxidation) and picked up by **NAD** or **FAD**, which are reduced.
- **Substrate-level phosphorylation**: a phosphate group is transferred directly to ADP, making ATP.

Link reaction (once per pyruvate, so twice per glucose)

1. Pyruvate (3C) is actively transported into the **matrix**.
2. It is **decarboxylated** (CO_2 removed) and **dehydrogenated** (hydrogen to NAD → reduced NAD). What's left is a 2C **acetyl group**.
3. The acetyl group combines with **coenzyme A** to form **acetyl CoA**, which carries it to the Krebs cycle.

- No ATP is made in the link reaction.

Krebs cycle (once per acetyl group, so twice per glucose)

1. The 2C acetyl group joins a **4C** compound to make a **6C** compound. Coenzyme A is released to collect another acetyl group.
2. The 6C compound is **decarboxylated** and **dehydrogenated**: a **5C** compound, CO_2 and reduced NAD are made.
3. The 5C compound is decarboxylated and dehydrogenated: a **4C** compound, CO_2 and reduced NAD.
4. The 4C compound is converted back into the original 4C acceptor. On the way, **1 ATP** is made by substrate-level phosphorylation, and **1 reduced FAD** and **1 more reduced NAD** are formed.

- It's a **cycle** because the 4C acceptor is regenerated, ready for the next acetyl group.
- Per turn: 2 CO_2, 1 ATP, 3 reduced NAD, 1 reduced FAD. Per glucose, double it.

Follow the carbon

6C glucose → 2 × 3C pyruvate → 2 × 2C acetyl + **2 CO_2** (link reaction) → **4 CO_2** (two turns of the Krebs cycle). That's 6 CO_2: every carbon atom of glucose has left as CO_2.

Tally per glucose

Stage	Where	CO_2	ATP (substrate-level)	Reduced NAD	Reduced FAD
Glycolysis	cytoplasm	0	2 (net)	2	0
Link reaction (× 2)	matrix	2	0	2	0
Krebs cycle (× 2)	matrix	4	2	6	2
Total		**6**	**4**	**10**	**2**

Most of the energy from glucose is now held in the **reduced coenzymes**. They take it to the electron transport chain (7.4).

EXAM TIP

Check whether a question means "per turn" or "per glucose". Per turn of the Krebs cycle: 2 CO_2, 1 ATP, 3 reduced NAD, 1 reduced FAD. Per glucose: twice that.

MARK-SCHEME WORDING

- Fate of pyruvate in aerobic respiration (any 3): "moves into the **matrix**"; "**dehydrogenated** / oxidised by NAD"; "**decarboxylated** / CO_2 removed" to "a **2C acetyl** group"; "combines with **coenzyme A** to form **acetyl CoA**"; "enters the **Krebs cycle**".
- Krebs cycle products (only the first three you list are marked): "carbon dioxide", "ATP", "reduced NAD", "reduced FAD". Numbers like "6 ATP" are ignored. Reduced NADP is not accepted.
- Drawing a mitochondrion: "**double membrane** with **cristae**", "**matrix** labelled as the site of the Krebs cycle".
- Data on isolated mitochondria: pyruvate is "used in the link reaction / aerobic respiration"; "**ADP** is needed to make ATP" (respiration slows if ADP runs out); an inhibitor of pyruvate transport "stops pyruvate entering the matrix". Refer to each sample by name (WBI15 Jun 2022).

Often asked as: MCQs on locations and on how substrates enter the Krebs cycle; name three products of the Krebs cycle; describe the fate of pyruvate; label the site of the Krebs cycle; magnification or width of a mitochondrion from an electron micrograph.

COMMON MISTAKE

- Putting the Krebs cycle on the cristae, or just "in the mitochondria". The link reaction and Krebs cycle are in the **matrix**; the electron transport chain is on the **inner membrane**.
- Mixing up decarboxylation (CO_2 removed) and dehydrogenation (hydrogen removed).
- Saying the link reaction makes ATP.
- Reciting the Krebs cycle in a data question without using the data.

⚠ Beyond the spec · not part of the Edexcel IAL specification

Examined in past papers even though the spec doesn't list it.

- Names the spec says you don't need are still credited: **acetyl CoA**, coenzyme A, **oxaloacetate** (the 4C acceptor), **citrate** (the 6C compound), "TCA cycle" (WBI15 Jan 2025).
- Other substrates: fatty acids are broken into 2C units that enter as acetyl CoA; amino acids enter as pyruvate, acetyl CoA or Krebs intermediates; glycerol enters glycolysis as triose phosphate.
- Magnification of a mitochondrion in standard form: convert mm to nm before dividing; watch the sign of the power of 10 (WBI15 Jan 2022, Jun 2022).

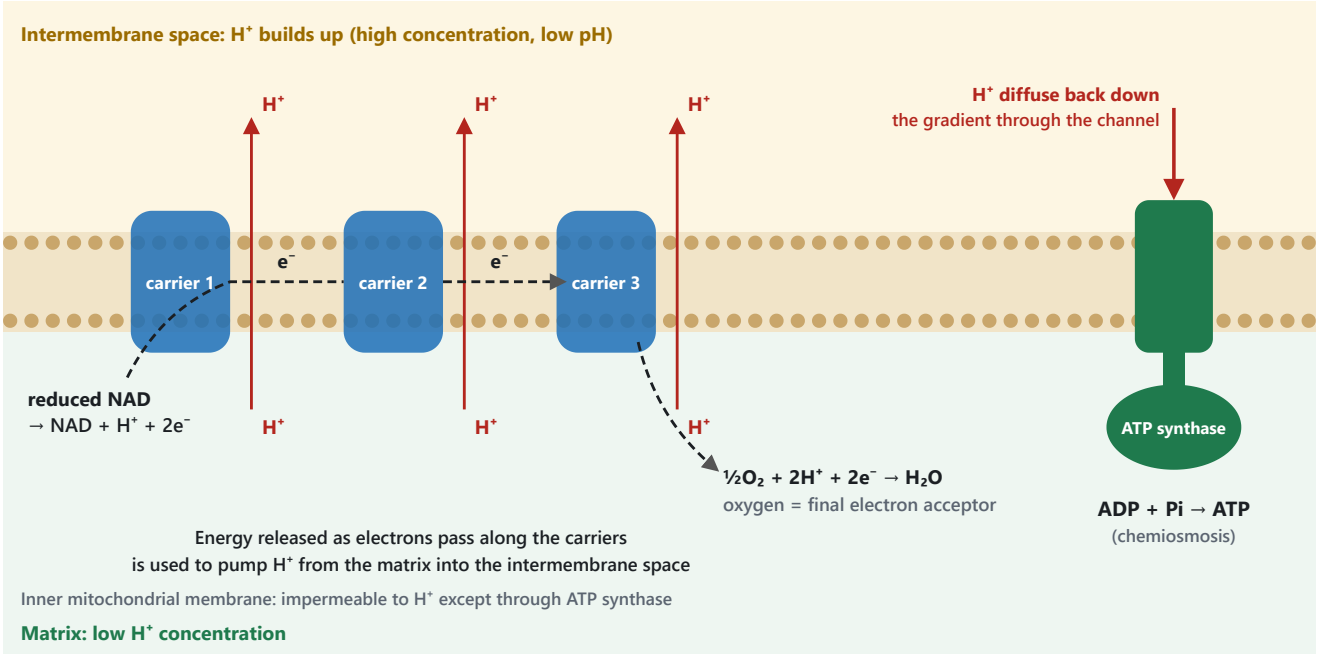
SPEC 7.4 Oxidative phosphorylation: the electron transport chain and chemiosmosis

WHAT THIS SPEC POINT ASKS

Understand how ATP is made by oxidative phosphorylation, linked to the electron transport chain in the mitochondria, including the roles of chemiosmosis and ATP synthase.

Step by step

1. Reduced NAD and reduced FAD deliver hydrogen atoms to the **electron transport chain (ETC)**: a series of electron carrier proteins in the **inner mitochondrial membrane** (cristae). The coenzymes are **reoxidised**, so NAD and FAD can return to the link reaction and Krebs cycle.
2. Each hydrogen atom splits into a **proton (H^+)** and an **electron (e^-)**.
3. The electrons pass from carrier to carrier in a series of redox reactions (each carrier is reduced, then oxidised). Each carrier is at a **lower energy level** than the one before, so **energy is released** at each step.
4. That energy is used to **pump (actively transport) H^+** from the **matrix** into the **intermembrane space**.
5. H^+ builds up in the intermembrane space: a **proton gradient** (an electrochemical gradient) forms across the inner membrane, which is impermeable to H^+.
6. H^+ **diffuses** back into the matrix down the gradient, but only through the channel in **ATP synthase**. This flow drives ATP synthase to join **ADP + Pi → ATP**. Making ATP using this flow of protons is **chemiosmosis**.
7. At the end of the chain, **oxygen is the final electron acceptor**. It combines with electrons and H^+ to form **water**: $\frac{1}{2}O_2 + 2H^+ + 2e^- \rightarrow H_2O$.



Electrons from reduced NAD pass along carriers in the inner membrane. The energy released pumps H^+ into the intermembrane space; H^+ diffuses back through ATP synthase, which makes ATP. Oxygen accepts the electrons at the end, forming water.

The first interactive diagram at the foot of the page runs the chain for you, so you can see what changes when oxygen runs out.

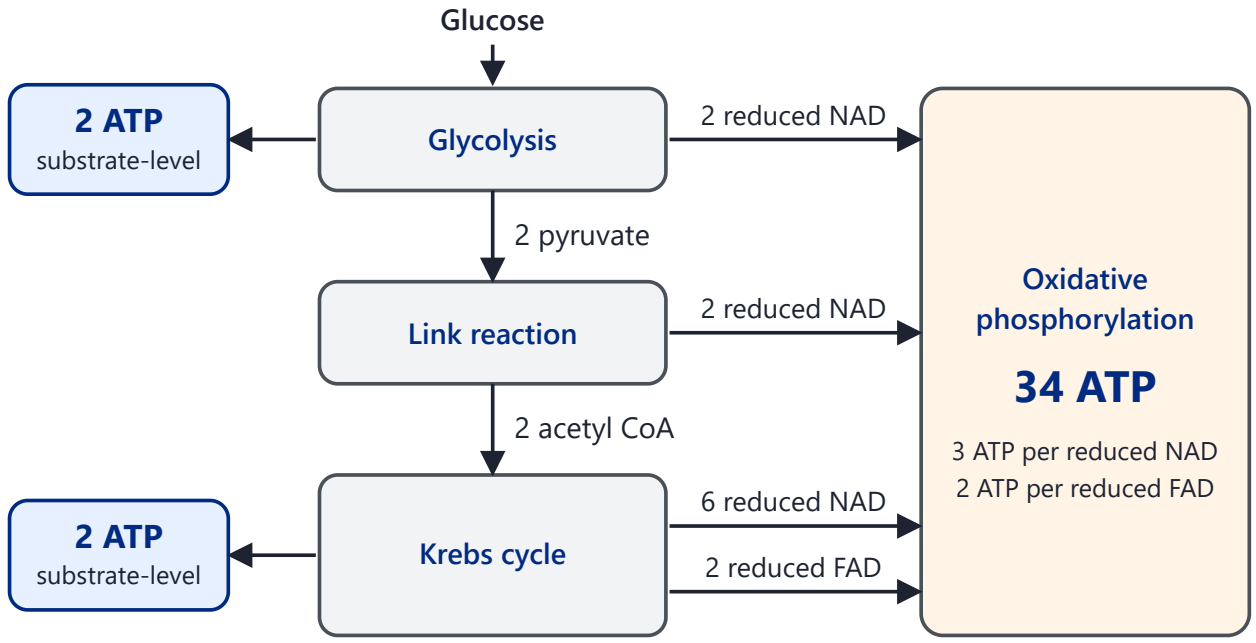
Why oxygen matters

- No oxygen → no final electron acceptor → electrons can't leave the last carrier → the chain stops → no H^+ is pumped → no chemiosmosis → very little ATP.
- Reduced NAD and reduced FAD can't be reoxidised, so the cell runs out of NAD and FAD. The link reaction and Krebs cycle stop too. Only glycolysis continues (anaerobic respiration).

How much ATP per glucose?

Source	Older textbook count	Modern estimate
Substrate-level phosphorylation (glycolysis + Krebs, net)	4	4
10 reduced NAD	10 × 3 = 30	10 × about 2.5 = 25
2 reduced FAD	2 × 2 = 4	2 × about 1.5 = 3
Total	**38**	**about 30–32**

- Real yields are lower than 38: some H^+ leaks back across the membrane without passing through ATP synthase, and energy is used to move pyruvate, ADP and Pi into the mitochondrion.
- Edexcel questions **give you the numbers**: 38 ATP and 30.51 kJ per ATP (WBI15 Jun 2021); 2 anaerobic v 36 aerobic (WBI15 Jun 2023). Use the values in the question.



Theoretical total: 2 + 2 + 34 = 38 ATP per glucose

ATP and reduced coenzymes produced at each stage, from one glucose molecule, using the older textbook count. Blue boxes are substrate-level ATP; the orange box is ATP from oxidative phosphorylation.

Theoretical ATP yield per glucose

1. Glycolysis: 2 ATP (substrate level) and 2 reduced NAD. Each reduced NAD gives 3 ATP, so 2 × 3 = 6.
2. Link reaction: 2 reduced NAD (one per pyruvate), so 2 × 3 = 6 ATP.
3. Krebs cycle (two turns): 2 ATP (substrate level), 6 reduced NAD (6 × 3 = 18 ATP) and 2 reduced FAD (2 × 2 = 4 ATP).
4. Total: 2 + 6 + 6 + 2 + 18 + 4 = 38 ATP.

This is the theoretical figure from the rules above. In practice the yield is lower, because some ATP is spent moving molecules across membranes, so check what your mark scheme expects.

MARK-SCHEME WORDING

- Chemiosmosis (any 5): "hydrogen (atoms) carried by **reduced NAD / reduced FAD** to the ETC"; "split into protons and electrons"; "**electrons** pass along the chain, **releasing energy**"; "energy used to **pump**

protons into the **intermembrane space**"; "protons **diffuse** back (down the electrochemical gradient) into the matrix through **ATP synthase**"; "forming ATP from ADP and Pi".

- Movement words: protons are "**pumped / actively transported**" out and "**diffuse**" back. "Stalked particle" is accepted for ATP synthase; "ATPase" is not. "ADP + Pi → ATP" with no mention of ATP synthase is ignored.
- Oxygen: "**terminal / final electron acceptor**", "accepts electrons and H^+ to form **water**", "no oxygen means electrons cannot pass along the chain".
- Cyanide (any 4): "electrons not passed on, so reduced coenzymes are not oxidised"; "cells respire **anaerobically** / glycolysis only"; "**less ATP** for muscle contraction"; "pyruvate converted to **lactate**"; "**pH falls**"; "enzymes denature".

Often asked as: describe the role of chemiosmosis in ATP synthesis (2–5 marks); describe oxidative phosphorylation; the role of reduced NAD or of oxygen; effects of inhibitors (cyanide, carbon monoxide, amytal, hydramethylnon); 6-mark interpretation of oxygen-consumption data for ETC defects.

COMMON MISTAKE

- Mixing up the active and passive steps: protons are **pumped** into the intermembrane space and **diffuse** back through ATP synthase.
- Confusing the **inner membrane** with the **intermembrane space**.
- Starting a sentence with "It…" so the examiner can't tell whether you mean NAD or reduced NAD. Name the molecule.
- "Affects ATP production." Say **increases** or **decreases**.
- Stopping at "anaerobic respiration" in an inhibitor question. Carry on: **less ATP**, so less muscle contraction (or whatever the context needs).

⚠ Beyond the spec · not part of the Edexcel IAL specification

Examined in past papers even though the spec doesn't list it.

- Named ETC complexes (I to IV), coenzyme Q and cytochrome c appear in data questions. You only need to follow where electrons enter and where a blocker stops them (WBI15 Jan 2021, 6-mark).
- **Carbon monoxide** binds to the haem in haemoglobin more strongly than O_2, so less O_2 reaches the cells to act as the final electron acceptor, and less ATP is made (WBI15 Jun 2023).
- An ETC inhibitor such as **amytal** reduces oxygen use and Krebs cycle activity, because NAD is no longer regenerated (WBI15 Jan 2025).
- Mitochondrial DNA codes for some ETC proteins, so mutations in it reduce ATP production (WBI15 Jan 2023 article).

SPEC 7.5 What happens to lactate after anaerobic respiration

WHAT THIS SPEC POINT ASKS

Understand what happens to the lactate made in animals after a period of anaerobic respiration.

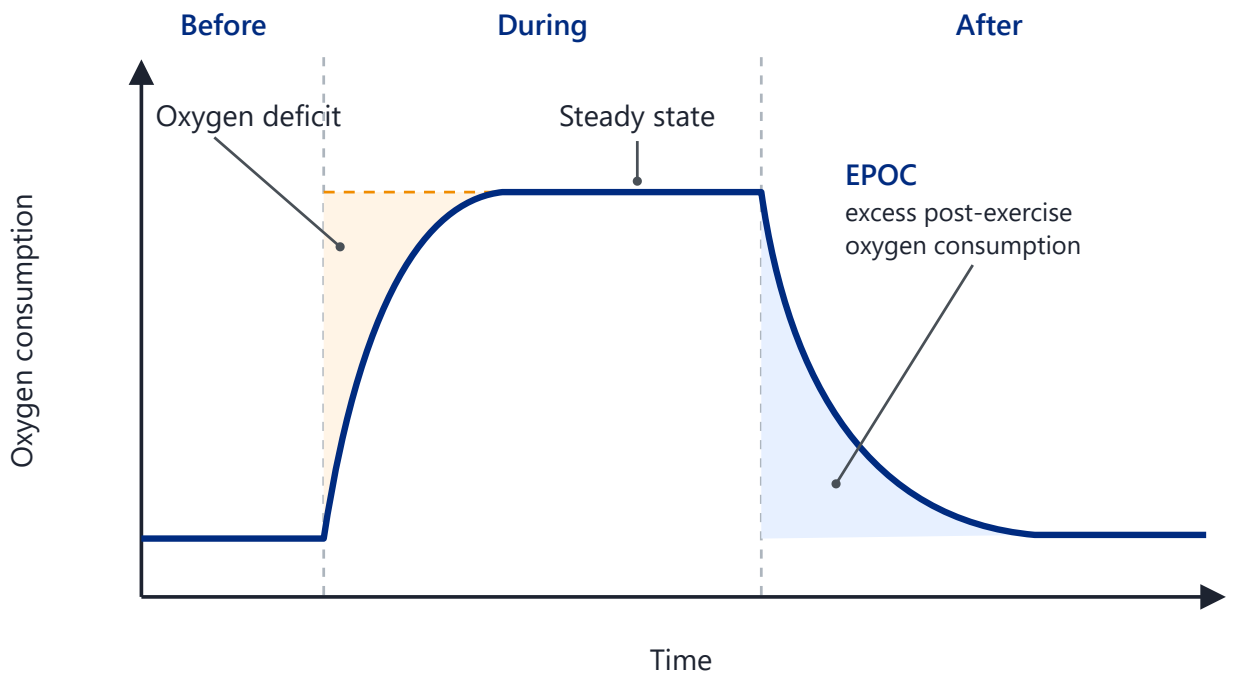
During hard exercise

- Muscles need ATP faster than oxygen can be delivered, so extra ATP comes from anaerobic respiration (glycolysis → lactate) and from creatine phosphate.
- Lactate builds up in the muscle and passes into the blood. Lactic acid releases H^+, so **blood pH falls**. Low pH inhibits enzymes and contributes to muscle **fatigue**.

After exercise

1. Lactate **diffuses into the blood** and is carried to the **liver** (heart muscle also takes some up and uses it).
2. Once oxygen is available, lactate is **oxidised back to pyruvate** (its hydrogen goes to NAD).
3. The pyruvate is then either:
    - **respired aerobically**: link reaction, Krebs cycle and ETC, giving CO_2, water and ATP; or
    - **converted to glucose** in the liver, which goes into the blood or is stored as **glycogen**.

- The extra oxygen needed for this (and to remake ATP and creatine phosphate and reload myoglobin with O_2) is the **oxygen debt**. That's why breathing rate and heart rate stay high for a while after exercise stops.



Oxygen consumption before, during and after exercise. The orange area is the oxygen deficit at the start of exercise; the blue area after exercise is the extra oxygen needed to recover (the oxygen debt, or excess post-exercise oxygen consumption).

WORKED EXAMPLE

Worked example: explaining a lactate graph

A cyclist's blood lactate rises over the first 15 minutes of a ride, then stays roughly constant from 15 to 60 minutes.

The question asks about 15 to 60 minutes only. Lactate is still being made by anaerobic respiration, but it is being **removed at the same rate** (converted back to pyruvate, or taken to the liver), so its concentration stays about the same (WBI15 Oct 2024). Don't explain the first 15 minutes too.

MARK-SCHEME WORDING

- Fate of lactate (any 2–3): "lactate **diffuses into the blood**"; "**transported to the liver**"; "**converted / oxidised back to pyruvate**" (ACCEPT dehydrogenation); "pyruvate then used in the **link reaction** / aerobic respiration"; "converted to **glucose or glycogen** in the liver".
- Lactate must become **pyruvate** first. "Lactate used in the link reaction / Krebs cycle" is ignored.
- "Aerobic respiration after the exercise repays the **oxygen debt**."
- Lactate concentration rises, so blood **pH falls**.

Often asked as: describe what happens to lactate at the end of a race; explain lactate graphs (cyclists, diving seals); compare the relative changes in lactate and heart rate; explain why breathing stays fast after exercise.

COMMON MISTAKE

- Leaving out the **liver**.
- Going down an "oxygen debt" route ("breathing removes lactate") instead of describing the **conversion** of lactate.
- Explaining the whole graph when the question names a time window.
- "An unfit person has more lactate." Compare **rates**: makes lactate sooner, or removes it more slowly.

⚠ Beyond the spec · not part of the Edexcel IAL specification

Examined in past papers even though the spec doesn't list it.

- Comparing two lines on different y-axes: use **percentage change**. Lactate rose by 308.5% while heart rate rose by only 22.5% (WBI15 Jan 2022).
- Seals diving: muscle lactate falls at the end of a dive as it moves into the blood and to the liver (WBI15 Oct 2025, 6-mark).

SPEC 7.6 Respiratory quotient (RQ)

WHAT THIS SPEC POINT ASKS

Understand what is meant by the respiratory quotient (RQ).

Definition

RQ = volume of CO_2 produced ÷ volume of O_2 used (over the same time). Moles can replace volumes, since equal numbers of moles of gas take up equal volumes at the same temperature and pressure.

Typical values (questions usually supply them)

Respiratory substrate	RQ
Carbohydrate (e.g. glucose)	1.0
Protein	about 0.8–0.9

Respiratory substrate	RQ
Lipid	about 0.7
Mixture of carbohydrate and lipid	between 0.7 and 1.0
Anaerobic respiration that releases CO_2 (yeast, plants)	above 1

Why lipids give a lower RQ

- Lipids contain proportionally **more hydrogen** and much less oxygen than carbohydrates.
- More O_2 is needed to turn all that hydrogen into water, so more O_2 is used for every CO_2 made: RQ is below 1.
- Glucose: $C_6H_{12}O_6 + 6O_2 \rightarrow 6CO_2 + 6H_2O$, so RQ = 6 ÷ 6 = **1.0**.

RQ for oleic acid

1. Equation for aerobic respiration of oleic acid: C18H34O2 + 25.5 O2 → 18 CO2 + 17 H2O + energy.
2. RQ = CO2 produced ÷ O2 used = 18 ÷ 25.5.
3. RQ = 0.7, which fits the value expected for a fat.

A fat needs far more oxygen per carbon than glucose, so less CO2 is released for each O2 used.

Reading RQ values

- **Above 1**: some anaerobic respiration that releases CO_2 (yeast, plants), or carbohydrate being converted into fat. Lactate fermentation in animals releases no CO_2 itself.
- **Falling over time**: the substrate is switching from carbohydrate to fat (e.g. a long cycle ride as glycogen stores run down).
- Measuring RQ needs **aerobic** respiration: the CO_2 comes from the link reaction and Krebs cycle, which only keep going when oxygen is there to act as the final electron acceptor (WBI15 Jan 2026).

WORKED EXAMPLE

Worked example: RQ from an equation and from data

From an equation: $2C_{18}H_{34}O_2 + 51O_2 \rightarrow 36CO_2 + 34H_2O$.

RQ = 36 ÷ 51 = **0.71**, so a lipid is being respired (WBI15 Jan 2024).

From measurements: a cockroach uses 1.7 cm^3 O_2 h^{-1} and produces 1.3 cm^3 CO_2 h^{-1}.

RQ = 1.3 ÷ 1.7 = 0.76 → **0.8** to 1 d.p. (WBI15 Oct 2021). Likely protein or a mixture of substrates.

Working backwards: O_2 uptake is 2.20 $dm^3\ min^{-1}$ and the RQ is 0.88.

CO_2 produced = RQ × O_2 used = 0.88 × 2.20 = **1.94 $dm^3\ min^{-1}$** (WBI15 Jan 2023).

MARK-SCHEME WORDING

- Definition (2 marks): "**carbon dioxide produced ÷ oxygen used / consumed**", as volumes or moles. "Amount" and "number of molecules" are ignored.
- Trend with a mixture: "as the proportion of lipid increases, the RQ decreases", supported by a calculation from the table.
- Cyclist: "RQ **decreased**" + "switching to **lipid / fat** rather than carbohydrate". Anaerobic respiration is ignored here.
- Why oxygen is needed to calculate RQ: "CO_2 is produced in the **link reaction / Krebs cycle**" and "oxygen is the **final electron acceptor** / needed for oxidative phosphorylation".

Often asked as: calculate RQ from an equation or gas volumes; work out CO_2 from the RQ and O_2 uptake; identify the substrate; interpret a changing RQ (a cyclist, germinating barley over 30 days).

COMMON MISTAKE

- Inverting it (O_2 ÷ CO_2).
- Describing the numbers but never naming the **substrate**.
- Rounding too early, or giving too many decimal places.
- "An RQ above 1 always means anaerobic respiration." It does only where anaerobic respiration releases CO_2 (yeast, plants); it can also mean fat is being made from carbohydrate.

⚠ Beyond the spec · not part of the Edexcel IAL specification

Examined in past papers even though the spec doesn't list it.

- Germinating barley over 30 days: carbohydrate is respired at first, lipid from about day 5 to day 10 (RQ falls), then the proportion of carbohydrate rises again. Describe each stage separately (WBI15 Jan 2026).
- A falling RQ during exercise means a **gradual change of substrate**; few candidates said this (WBI15 Jan 2023).

SPEC 7.7 Core Practical 15: a redox indicator and respiration in yeast

WHAT THIS SPEC POINT ASKS

Use an artificial hydrogen carrier (a redox indicator) to investigate respiration in yeast.

Principle

- **Dehydrogenase** enzymes in glycolysis and the Krebs cycle remove hydrogen from respiratory substrates. Normally NAD (or FAD) accepts it.
- A **redox indicator** is an **artificial hydrogen acceptor**: it takes the hydrogen instead, and **changes colour when it is reduced**.
  - methylene blue: blue → colourless
  - DCPIP: blue → colourless
  - TTC (triphenyltetrazolium chloride): colourless → red
- The faster the yeast respires, the faster the colour changes. **Rate = 1 ÷ time** for the colour change.

CORE PRACTICAL

Core practical 15: effect of temperature on respiration in yeast

1. Make a suspension of yeast in glucose solution (activated in warm glucose beforehand). Use the same volume and concentration in every tube.
2. Put tubes of yeast suspension, and **separate** tubes of indicator (e.g. 1 cm^3 of methylene blue), in **thermostatically controlled water baths** at a range of temperatures, e.g. 20, 25, 30, 35 and 40 °C. Leave them 10 minutes to **equilibrate**.
3. Add the indicator to the yeast, mix once by inverting, and start the timer.
4. **Don't shake again**: oxygen from the air would re-oxidise the indicator. (A thin layer of oil on top keeps air out.)
5. Stop the timer when the blue colour has gone, judged against a **colour standard** (yeast suspension with water in place of the indicator).
6. Repeat three times at each temperature. Calculate the mean time and the rate (1 ÷ mean time).

Controls: boiled yeast + indicator (no colour change: shows living yeast and active enzymes are needed); yeast with no glucose (much slower).

Variables

- **Independent:** temperature (or glucose concentration, or type of sugar).
- **Dependent:** time taken for the indicator to change colour, converted to a rate (s^{-1}).

- **Control:** volume and concentration of yeast suspension; volume and concentration of glucose; volume and concentration of indicator; pH (use a buffer); how the tube is mixed; same person judging the end point.

WORKED EXAMPLE

Worked example: rate from time

Mean time for methylene blue to lose its colour: **250 s** at 25 °C and **125 s** at 35 °C.

Rate at 25 °C = 1 ÷ 250 = **4.0×10^{-3} s^{-1}**. Rate at 35 °C = 1 ÷ 125 = **8.0×10^{-3} s^{-1}**.

The rate doubles for a 10 °C rise: molecules have more kinetic energy, so more enzyme–substrate complexes form. Above the optimum, enzymes denature (active sites change shape) and the rate falls.

Errors and improvements

- The end point is **subjective** (yeast is cream-coloured) → use a colour standard and the same observer, or a **colorimeter** reading absorbance at set times.
- Indicator re-oxidised by air → don't shake; oil layer.
- Yeast settles → stir the stock suspension before taking each sample.
- Safety: methylene blue stains (gloves); take care with hot water baths.

MARK-SCHEME WORDING

- Applied version, the MTT assay (MTT turns yellow → purple when reduced): "use **living / viable** cells"; "**reduced NAD** is produced in respiration and reduces the dye"; "same concentration of each drug; same number of cells and conditions"; "record the **time** for the colour to change (or use a colorimeter)"; "the **longer** it takes to turn purple, the **more toxic** the drug" (WBI15 Jan 2022 article).
- Planning language (from Unit 6): "range of five temperatures", "**thermostatic** water bath", "equilibrate yeast and indicator **separately**", "named indicator", "add **glucose** (respiratory substrate)", "repeats and means".

Often asked as: not set as a method question in WBI15 from 2020 to 2026; its logic appears in unfamiliar contexts such as the MTT toxicity assay. Unit 6 asks for the full plan.

COMMON MISTAKE

- Forgetting the glucose, or "thermostatically controlled" for the water bath. Shaking the tube while timing.
- Reversing the conclusion: a drug that slows respiration slows the colour change, so a **longer** time means a **more** toxic drug. Few candidates mentioned using living cells.

SPEC 7.8 Core Practical 16: respirometer, rate of respiration and

RQ

WHAT THIS SPEC POINT ASKS

Use a simple respirometer to find the rate of respiration and the RQ of a suitable material, such as germinating seeds or small invertebrates.

Principle

- Organisms in a sealed tube take in O_2 and give out CO_2.
- **Soda lime** (or potassium hydroxide) **absorbs the CO_2**, so the only change in gas volume is the **O_2 taken in**. The gas volume and pressure fall, and the coloured liquid in the capillary tube moves **towards** the organisms.
- Volume of O_2 = distance moved × cross-sectional area of the capillary = **$\pi r^2 l$** (r = radius of the bore, l = distance moved).
- A **control tube** contains **glass beads** of the same volume as the organisms (and the same soda lime). Any movement in it is caused by changes in temperature or pressure, so it can be used to correct the readings.

CORE PRACTICAL

Core practical 16: rate of respiration and RQ

1. Weigh the organisms (e.g. 2.0 g of germinating seeds, or woodlice or maggots). Put soda lime in the tube, a **wire gauze** above it, then the organisms on the gauze.
2. Set up the control tube with glass beads of equal volume.
3. Stand both tubes in a **thermostatically controlled water bath** (e.g. 25 °C) with the tap open for 10 minutes to **equilibrate**.
4. Close the tap. Mark the liquid's position and measure the **distance it moves in a set time** (e.g. 5 minutes). Reset it with the syringe.
5. Repeat at least three times and calculate the mean. Rate = $\pi r^2 l$ ÷ time (÷ mass for a rate per gram).
6. **For the RQ:** repeat with the **same organisms**, but with the soda lime replaced by the same volume of glass beads, so CO_2 isn't absorbed.

The second interactive diagram at the foot of the page lets you do both runs and work out the RQ.

Getting the RQ from the two runs

- With soda lime, movement **x** = O_2 taken in.
- Without soda lime, movement **y** = O_2 taken in − CO_2 given out (the net change).
- So CO_2 given out = x − y, and **RQ = (x − y) ÷ x**.
- If the liquid doesn't move without soda lime, CO_2 out = O_2 in, so RQ = 1 (carbohydrate).

- If it moves **away** from the organisms without soda lime, more CO_2 is given out than O_2 taken in: add that distance instead of subtracting it (RQ above 1).

WORKED EXAMPLE

Worked example: maggots

Capillary radius 0.5 mm; 1.5 g of maggots; 25 °C; 5 minutes per reading.

With soda lime, the liquid moves **25 mm** towards the maggots. Without soda lime, it moves **5 mm** towards them.

O_2 used = $\pi r^2 l$ = 3.14 × 0.5² × 25 = **19.6 mm³** in 5 min → 19.6 ÷ 5 = **3.9 mm³ min⁻¹** → 3.9 ÷ 1.5 = **2.6 mm³ g⁻¹ min⁻¹**.

CO_2 produced = 25 − 5 = 20 mm of capillary. RQ = 20 ÷ 25 = **0.80**, so protein (or a mixture of substrates) is being respired.

The same capillary is used for both runs, so πr^2 cancels in the RQ and you can use the distances directly.

Control variables, safety and errors

- **Control:** mass and type of organism; temperature; volume of soda lime; capillary bore; equilibration time; age of the seeds or stage of the larvae.
- **Safety and ethics:** soda lime and KOH are **corrosive** (gloves, eye protection); the gauze stops organisms touching them; handle invertebrates gently, keep runs short and return them to their habitat.
- **Errors:** leaks at the bung (seal with petroleum jelly); temperature changes (water bath and control tube); soda lime becoming saturated (use fresh); small movements are hard to read (finer capillary, longer time, or a gas syringe or pressure sensor); seeds may switch to anaerobic respiration as O_2 runs low (keep runs short).

MARK-SCHEME WORDING

- "CO_2 absorbed by **soda lime / KOH**, so movement shows **O_2 uptake**". Sodium hydrogencarbonate is **not** a CO_2 absorber.
- "**Control tube** with glass beads of the same volume" corrects for temperature and pressure changes.
- Glass beads replace soda lime in the second run "**so carbon dioxide is not absorbed**" (and the volume stays the same).
- RQ = "volume of CO_2 produced ÷ volume of O_2 used".
- Rate answers need the unit conversions done (cm³ to dm³, minutes to hours) and the s.f. or standard form asked for.

Often asked as: not set as a method question in WBI15 from 2020 to 2026; instead, RQ and rate calculations from given values (WBI15 Oct 2021, Jan 2023, Jan 2026). The full plan is a favourite in Unit 6.

COMMON MISTAKE

- Saying a respirometer measures CO_2 output. It measures **O_2 uptake**; the CO_2 comes from the difference between the two runs.
- Using the **diameter** instead of the radius in πr^2.
- A control tube with soda lime only. It needs **inert glass beads** of the same volume as the organisms.

Quick checks

QUICK CHECK

In which part of the cell does the Krebs cycle take place?

- Cytoplasm
- Mitochondrial matrix
- Intermembrane space
- Cristae of the inner membrane

The Krebs cycle runs in the matrix. The cytoplasm is where glycolysis happens, which is the commonest mix-up. The cristae hold the electron transport chain, not the cycle.

QUICK CHECK

What is the net number of ATP produced by glycolysis per glucose molecule?

- 0
- 2
- 4
- 32

Glycolysis makes 4 ATP by substrate-level phosphorylation but uses 2 at the start to phosphorylate glucose, so the net gain is 2. Quoting 4 is the common error of forgetting the ATP spent.

QUICK CHECK

Why does the electron transport chain stop when oxygen is absent?

- Oxygen is needed to split glucose into pyruvate
- No final electron acceptor, so the electrons and H+ cannot be removed from the chain
- Oxygen is the substrate for ATP synthase
- Lactate blocks the carriers

Oxygen accepts electrons at the end of the chain and combines with H+ to form water. Without it, the carriers stay reduced and reduced NAD cannot be reoxidised, so the chain and chemiosmosis stop.

QUICK CHECK

Why is lactate formed in muscle during anaerobic respiration?

- To store energy as ATP for later use
- To remove carbon dioxide from the cell
- To regenerate NAD+ so that glycolysis can continue
- To pump H+ into the intermembrane space

Pyruvate accepts hydrogen from reduced NAD, which regenerates NAD+ for glycolysis. Lactate itself makes no extra ATP, so the first option is wrong.